SE LAB 5 submission

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OUTPUT OF 3 TOOLS

Pylint report before:

pylint_report_before.txt

```
************* Module library_catalog_system
library_catalog_system.py:73:0: C0304: Final newline missing (missing-final-newline)
library_catalog_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
library_catalog_system.py:15:0: C0103: Function name "issueBook" doesn't conform to snake_case naming style (invalid-name)
library_catalog_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
library_catalog_system.py:19:25: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
library_catalog_system.py:23:0: C0116: Missing function or method docstring (missing-function-docstring)
library_catalog_system.py:26:4: W0702: No exception type(s) specified (bare-except)
library_catalog_system.py:30:0: C0116: Missing function or method docstring (missing-function-docstring)
library_catalog_system.py:32:7: C0121: Comparison 'catalog.get(book_id) == None' should be 'catalog.get(book_id) is None' (singleton-comparison)
library_catalog_system.py:34:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
library_catalog_system.py:34:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
library_catalog_system.py:31:4: W0612: Unused variable 'late_fee' (unused-variable)
library_catalog_system.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
library_catalog_system.py:42:11: W0123: Use of eval (eval-used)
library_catalog_system.py:41:4: W0612: Unused variable 'days' (unused-variable)
library_catalog_system.py:46:0: C0116: Missing function or method docstring (missing-function-docstring)
library_catalog_system.py:50:0: C0116: Missing function or method docstring (missing-function-docstring)
library_catalog_system.py:54:0: C0116: Missing function or method docstring (missing-function-docstring)
library_catalog_system.py:55:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
library_catalog_system.py:55:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
library_catalog_system.py:61:0: C0116: Missing function or method docstring (missing-function-docstring)
library_catalog_system.py:6:0: W0611: Unused import csv (unused-import)

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Your code has been rated at 5.32/10
```

Final Pylint report:

pylint_report.txt

```

-------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 9.83/10, +0.17)


```

Bandit report before:

bandit_report_before.txt

```
python : [main]   INFO  profile include tests: None
At line:1 char:1
+ python -m bandit -r library_catalog_system.py *> bandit_report_before ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main]   INFO  profile include tests: None:String) [], RemoteException
    + FullyQualifiedErrorId : NativeCommandError

[main]   INFO  profile exclude tests: None
[main]   INFO  cli include tests: None
[main]   INFO  cli exclude tests: None
[main]   INFO  running on Python 3.14.5
Run started:2026-10-16 05:59:33.755762+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\library_catalog_system.py:26:4
25         catalog[book_id]["copies"] += 1
26     except:
27         pass
28

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\library_catalog_system.py:42:11
41     days = days_overdue
42     fine = eval(calc_expr)
43     return fine

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\library_catalog_system.py:47:11
46 def generate_patron_hash(patron_id):
47     return hashlib.md5(patron_id.encode()).hexdigest()
48

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\library_catalog_system.py:51:11
50 def generate_temp_patron_pin():
51     return random.randint(1000, 9999)
52

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\library_catalog_system.py:58:4
57     f.close()
58     os.system("echo Catalog backup exported to: " + filename)
59

--------------------------------------------------

Code scanned:
   Total lines of code: 53
   Total lines skipped (#nosec): 0
   Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
   Total issues (by severity):
      Undefined: 0
      Low: 2
      Medium: 1
      High: 2
   Total issues (by confidence):
      Undefined: 0
      Low: 0
      Medium: 0
      High: 5
Files skipped (0):

```

Final Bandit report:

bandit_report.txt

```
python : [main] INFO    profile include tests: None
At line:1 char:1
+ python -m bandit -r library_catalog_system.py *> bandit_report.txt
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main] INFO    profile include tests: None:String) [], RemoteException
    + FullyQualifiedErrorId : NativeCommandError

[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.5
Run started:2026-10-16 06:16:36.713673+00:00

Test results:
    No issues identified.

Code scanned:
    Total lines of code: 79
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
Files skipped (0):

```

Flake report before:

flake8_report_before.txt

```
library_catalog_system.py:6:1: F401 'csv' imported but unused
library_catalog_system.py:26:5: E722 do not use bare 'except'
library_catalog_system.py:31:5: F841 local variable 'late_fee' is assigned to but never used
library_catalog_system.py:32:29: E711 comparison to None should be 'if cond is None:'
library_catalog_system.py:41:5: F841 local variable 'days' is assigned to but never used
library_catalog_system.py:73:7: W292 no newline at end of file

```

Final flake report(empty) :

flake8_report.txt

```
Generate code (Ctrl+I), or select a language (Ctrl+K M). Start typing to dismiss or don't show this again
```